

Rahul Sharma - Work Sample Portfolio

Prepared for hiring managers and technical interview loops.

Profile: Data Scientist / AI Engineer

Experience: 5+ years in production ML, GenAI systems, and applied analytics

Core themes: Product-focused AI systems, agentic workflows, RAG, multimodal pipelines, predictive modeling, MLOps, cloud deployment, and analytics-to-decision delivery

Portfolio breadth: 12+ projects spanning healthcare AI, operations intelligence, learning systems, and multimodal ML

Report Scope and Portfolio Context

- This report highlights three flagship hosted systems: AI Cargo, Doc2Data, and SkillSpring.
- These projects were selected to represent applied AI delivery across operations, healthcare, and learning products.
- Additional projects are available in the portfolio repository, including Streamlit-hosted prototypes and research-focused implementations.
- The structure below is designed for recruiter and hiring-manager scanability: problem, scope, architecture, links, and impact.

Project 01 - AI Cargo

- Agentic Operations
- Cold-chain Risk
- Compliance AI

Overview

AI Cargo is an agentic risk intelligence platform for pharmaceutical cold-chain shipments. It combines deterministic compliance rules and ML predictions, then orchestrates mitigation workflows through tool-using agents with human-in-the-loop controls.

Problem Solved

Temperature-sensitive logistics requires immediate, compliant, and auditable decisions. Manual intervention is often slow and inconsistent for medium/high risk events.

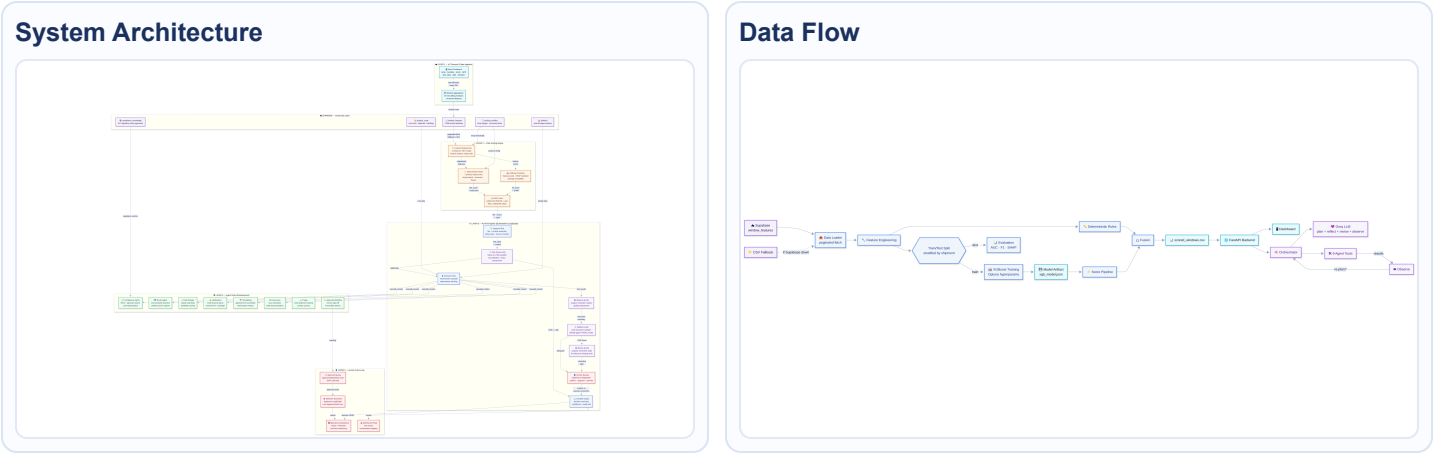
End-to-End System Scope

1. Ingest telemetry windows from Supabase with fallback data support.
2. Engineer risk features and score windows via deterministic + XGBoost fusion.
3. Route medium/high/critical events to LangGraph orchestration.
4. Execute specialized tools (compliance, routing, scheduling, insurance, triage, approvals, notifications).
5. Surface outcomes in FastAPI + React dashboard with traceability and audit logs.

Architecture Diagram

System architecture: [View link](#)

Data flow architecture: [View link](#)



Demo

Live demo: [Open application](#)

GitHub

Source code: [View repository](#)

Tech Stack

Python, FastAPI, LangGraph, XGBoost, SHAP, Optuna, Supabase, React, Tailwind.

Business Impact Narrative: Improves intervention speed for high-risk cold-chain incidents, adds compliance transparency through RAG-backed validation and logging, and demonstrates practical agentic orchestration with governed human approval.

Project 02 - Doc2Data

Multimodal AI

Document Intelligence

Healthcare Automation

Overview

Doc2Data is a healthcare document-to-JSON extraction platform that processes CMS-1500 and UB-04 forms using lane-based extraction, alignment, OCR, validation, and targeted rescue logic.

Problem Solved

Healthcare form extraction is noisy and expensive when every document is forced through the same OCR pipeline. Doc2Data adapts extraction strategy per document type and quality signal.

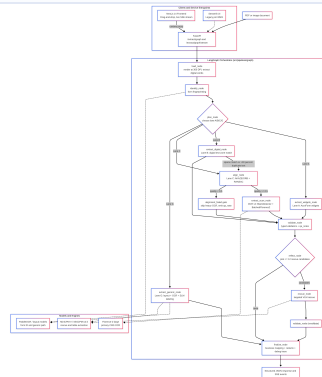
End-to-End System Scope

1. Upload PDF/image through Next.js frontend.
2. LangGraph identifies form type and chooses extraction lane (widgets, digital text, scan OCR, generic).
3. Alignment + OCR pipeline executes with quality gates.
4. Validation pass identifies uncertain fields.
5. Targeted VLM rescue runs only on selected fields.
6. Final output returns structured fields, metadata, validation notes, and debug traces.

Architecture Diagram

System architecture: [View link](#)

System Architecture



Demo

Live demo: [Open application](#)

GitHub

Source code: [View repository](#)

Tech Stack

FastAPI, Next.js, LangGraph, Florence-2 OCR, VLM rescue, Docker.

Business Impact Narrative: Reduces extraction time from multi-minute runs to production-viable latency, improves reliability with lane routing, validation, and confidence-aware rescue, and produces integration-ready JSON for downstream healthcare workflows.

Full-stack GenAI Learning Intelligence Career Planning

Comparative Strength Across Work Samples

- **SkillsSpring:** Product-oriented full-stack GenAI with learning + career intelligence.
- **AI Cargo:** Agentic operations and compliance under real-world risk constraints.
- **Doc2Data:** Multimodal document AI with robust extraction and quality controls.

Together, these projects show system design depth across user-facing UX, AI orchestration, data pipelines, and production delivery.

Contact

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